Nguyen Duc An

Backend Software Engineer

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About Me

I am a Backend Software Engineer with 3 years of experience, including 1 year at Axon and over 2 years at Zalopay. A high-performing engineer who embraces challenges, welcomes feedback, and continuously strives for excellence through learning and collaboration to deliver meaningful impact for users and businesses.

Education

Bachelor of Computer Science - Honors Program Sep 2020 – Sep 2024
Ho Chi Minh City University of Technology (HCMUT)

- GPA: 3.85/4.0
- Graduation classification: Very Good.
- Major: **Honors Program** - Software Engineer.

Experience

Software Engineer I, Axon, Vietnam Mar 2026 – Present

- Working on the **Integration Team** within Axon Records Management System (RMS), developing a **Data Platform** that orchestrates data ingestion and egress pipelines to enable reliable data exchange between RMS and external systems.
- **Submission State Machine**: Designed and implemented a submission state machine for the orchestrator service of the Generic Compliance Module (GCM) to manage police report submission lifecycles across external state and federal systems, with around **4M+ submissions**.
  - Implemented **cursor-based pagination** for endpoint **TailSubmissions** and worker concurrency controls for batch processing. Applied database-level concurrency controls and idempotency mechanisms to prevent duplicate processing while handling approximately **6,000 submissions daily**.
  - Implemented worker workflows and activities for batched historical status backfill using the internal **Temporal** platform. Applied the **Expand–Migrate–Contract** pattern and **feature flags** to execute a **backward-compatible**, **zero-downtime** migration.
- **Generic Compliance Module (GCM)**: Implemented a Submission Tab in the Manual Submission Module for **20M+ reports** across **67 active agencies**, covering **10 compliance report types** and **26 U.S. state portals**. Supported submission creation, status tracking, error visibility, and recall/resubmission.
- **Compliance Ingestion Endpoint**: Developed a generic ingestion endpoint for multiple compliance report types, capturing DB changes from the CDC pipeline (**Debezium CDC, Kafka, consumer**) to propagate document finalization events from the document service to the submission service. Implemented report-type validation, compliance-system routing, and an idempotent mechanism to ensure reliable ingestion.
- **UCAD Integration:** Implemented a poll-based scheduler, ingestion endpoint, and Liquid templates via the Integration Framework to ingest and transform UCAD incident data into RMS. Implemented resilient retries, backpressure with jitters, and queue-depth alerts. Validated scalability through load testing across **100 agencies**, achieving **300 ms p95 batch latency** for batches of **100 incidents**.

Middle Software Engineer, Zalopay – VNG, Vietnam Aug 2025 – Mar 2026

- **Tracking Advertising Hub**: Designed and developed a high-throughput user behavior tracking platform capable of handling **10,000+** RPS, integrating with advertising conversion APIs (Firebase GA4, Appsflyer, TikTok, Meta) for attribution and analytics. Designed a Kafka-based event pipeline to ingest and stream events in real time and

applied **Worker Pool patterns** and concurrency programming in Golang with channels and goroutines.

- **Live Activities (Tet Campaign)**: Built a distributed Go notification fan-out system for Tet 2026 countdown campaigns reaching **2M+ iOS users**, leveraging Kafka and concurrent workers for high-volume APNS delivery with rate limiting and idempotency.

Associate Software Engineer, Zalopay – VNG, Vietnam Apr 2024 – Aug 2025

- **Zalopay Onelink Solution**: Designed and built the Onelink system from scratch in Go for 200+ mini-apps, enabling accurate deep linking across **1,000+** social campaigns. Optimized **redirect latency** to **under 10 ms** using Redis caching strategies. Integrated Kafka pipelines for high-throughput tracking of clicks, redirects, and drop rates in analytics dashboards.
- **A/B Testing Platform**: Developed a scalable experimentation platform with feature flags and traffic segmentation, using Weighted Round-Robin scheduling to distribute users based on configurable rollout percentages (e.g., 1%, 25%, 50%, 100%), supporting controlled experimentation and gradual capacity expansion across variants.

Fresher Software Development, Zalopay – VNG, Vietnam Jan 2024 – Apr 2024

- Attended ZaloPay Annual Fresher Training Camp, one of 15 top Freshers out of 3,000+ applicants.
- **Mini App Platform:** Designed and maintained high-performance RESTful and gRPC APIs in Go for a mini-app platform serving **200+ internal and merchant mini apps**. Built an internal management platform to configure mini-app data and enforce RBAC-based access control.

Technologies

- **Languages & Frameworks**: Go, Java, JavaScript, TypeScript, React.
- **Backend & System Design**: Microservice Patterns, Problem Analysis, Solution Design, Scope Framing.
- **Infrastructure**: MySQL (database), Redis (caching), Kafka (message queue), CDN.
- **Observability**: Prometheus, Grafana, Splunk.
- **Background Knowledge**: Data Structures & Algorithms, Operating Systems, Networking.

English Proficiency & Certificates

- **TOEIC**: Listening & Reading 860/990; Speaking & Writing 300/400.

Awards

Top 2 Zalopay Hackathon - AI Solutions - Scan Everything Sep 2025

- Developed an AI-driven "Scan Everything" solution that enables users to perform image-based searches across the company's mini-app ecosystem. The system recommends Top-K relevant mini apps and items, unlocking opportunities for discovery and cross-selling mini apps.

Top 8 NAB AWS Security Jam 2023 - The JamGuardians Oct 2023

- We address security-related issues within AWS by utilizing AWS services, ranking among the top 8 out of over 30 teams. Tech stack: EC2, S3, Lambda, AWS KMS, AWS IAM, AWS VPC.

Top 4 GDSC Hackathon Vietnam 2023 - Pollunite Mar 2023

- Out of 110 registration forms and 50 project ideas, we were proud to receive the Consolation Prize at the hackathon for our environmental initiative—raising awareness about littering and creating a system to reward local residents for collecting trash through community-driven fundraising.